

360° Education Business Analysis

Prepared By

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Tools Used: Python, MySQL, Power BI

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1. Project Overview

The 360° Education Business Analysis project is a comprehensive data-driven evaluation designed to optimize the operational, financial, and placement ecosystems of a multi-branch professional training institute. Over a rolling 17-month period (January 2024 – May 2025), the project consolidates performance metrics across five geographic branches (Karve Nagar, Nagpur, Hadapsar, Pimpri, and Chinchwad) and five core technology disciplines (Data Science, Data Analytics, Java Full Stack, Python Full Stack, and Software Testing).

By analyzing the critical intersections between marketing lead generation, academic admissions, financial collections, and corporate placement outcomes, this project provides executives with a single source of truth to streamline operations, plug revenue leakage, and improve student career success.

2. Problem Statement

Despite generating substantial top-line revenue ($\$66\text{M}$) and maintaining a steady influx of student enrollments ($\$2\text{K}$ admissions), the institute suffers from critical structural inefficiencies that threaten its long-term financial health and brand reputation:

- **Severe Cash Flow Stagnation:** Out of $\$66\text{M}$ in total booked revenue, **$\$14\text{M}$ (21.2%) remains uncollected** as pending fees. Exactly 50% ($\$1\text{K}$ out of $\$2\text{K}$) of the total student base has outstanding balances, indicating an ineffective or overly lenient fee collection mechanism.
- **Underperforming Placement Pipeline:** The institute's core value proposition is career transition, yet the overall placement rate stands at a low **34.85%** ($\$697$ students placed out of $\$2\text{K}$). Despite corporate goodwill and the generation of over $\$6,000$ interview calls, the conversion from interview to corporate hire is bottlenecked.

- **Extreme Operational Volatility:** The business experienced a sudden, catastrophic collapse in revenue, admissions, and placement metrics in **May 2025**. This signals either an unmitigated operational failure (e.g., severe seasonal drops, sudden market irrelevance) or major data pipeline/logging gaps.
- **Sub-optimal Marketing Spend:** While high-intent channels like LinkedIn, YouTube, and physical Walk-Ins drive the highest conversion numbers, marketing efforts remain split across less effective platforms, capping the efficiency of the lead-to-admission pipeline (\$31.9\%\$ conversion rate).

3. Objectives

- **Recover Outstanding Revenue:** Reduce the **14M** in pending fees and boost fee collection efficiency from **79.45% to over 95%** using stricter milestone-locked payment policies.
- **Boost Placement Rates:** Elevate the institutional placement rate from a low **34.85% to 65%** by introducing mandatory mock interviews and better technical screening preparation.
- **Maximize Marketing ROI:** Increase the sales conversion rate from **31.90% to 40%** by shifting ad spend away from low-performing channels and heavily into **LinkedIn and YouTube**.
- **Fix Data & Operational Anomaly:** Investigate and resolve the drastic performance drop-off in **May 2025** while correcting dashboard layout/labeling errors to ensure accurate executive reporting.

4. Tools & Technologies Used

This project was developed using visualization tools:

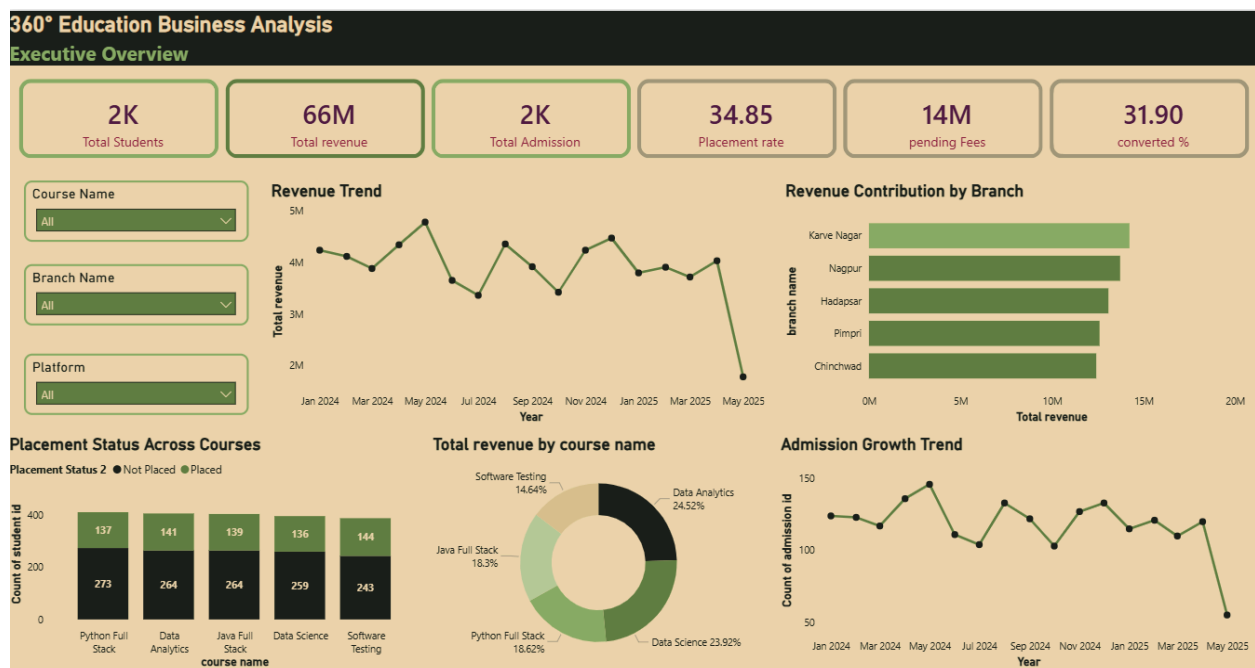
- Microsoft Power BI → Interactive dashboards, KPIs, and reporting

5. Power BI Dashboard Development

5.1 Executive Overview

This dashboard acts as the high-level health check for the business, combining sales, finance, and operational metrics.

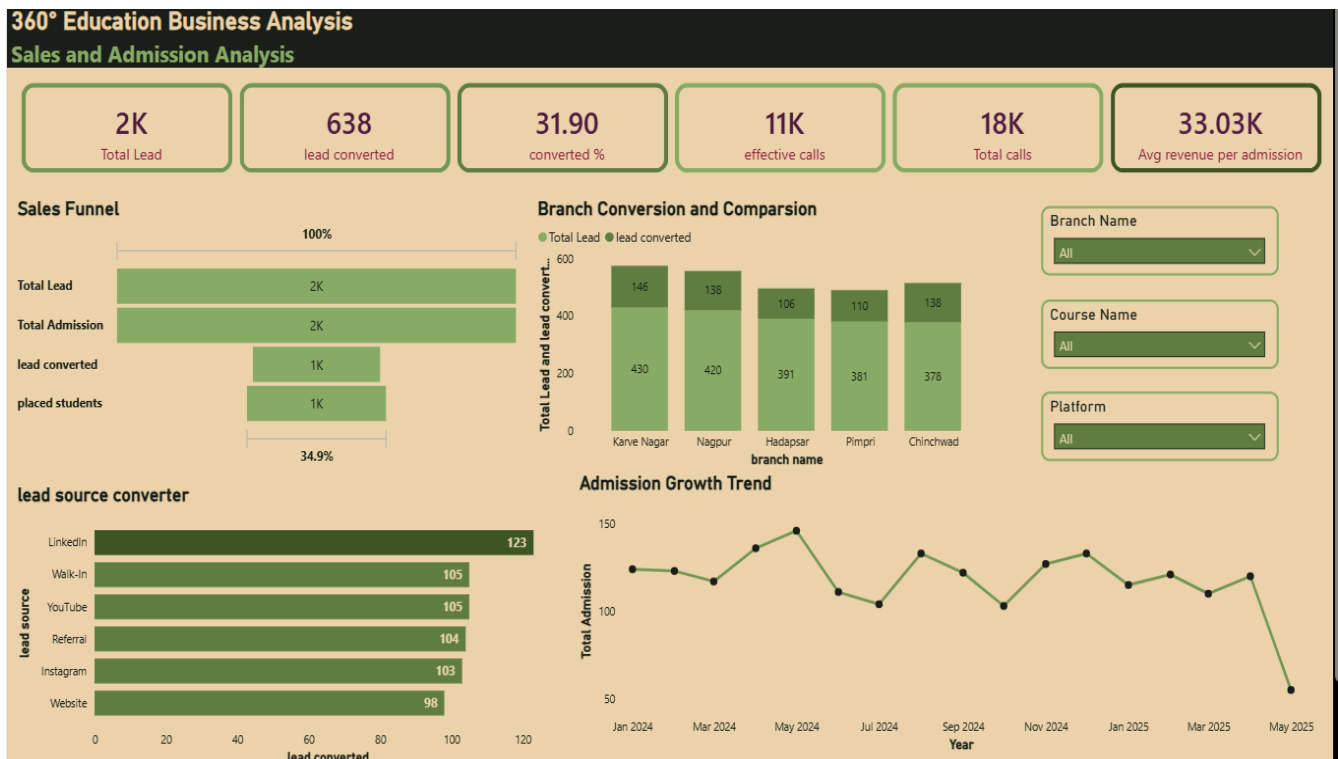
- **Key Metrics (KPIs):** Total Students (2K), Total Revenue (66M), Total Admission (2K), Placement Rate (34.85%), Pending Fees (14M), and Conversion Rate (31.90%).
- **Visualizations & Structure:**
 - **Revenue Trend:** Highlights a highly volatile revenue graph that peaked in May 2024 but suffered a massive cliff-dive drop in May 2025.
 - **Revenue Contribution by Branch:** Shows a relatively balanced performance across branches, with Karve Nagar leading (~14.2M) and Chinchwad at the bottom (~12.4M).
 - **Total Revenue by Course:** Data Analytics (24.52%) and Data Science (23.92%) drive nearly half of all revenue.
 - **Placement Status Across Courses:** Stacked bar chart showing that across all courses, the number of "Not Placed" students heavily outweighs the "Placed" students.



5.2 Sales and Admission Analysis

This dashboard tracks the efficiency of the marketing and sales pipeline.

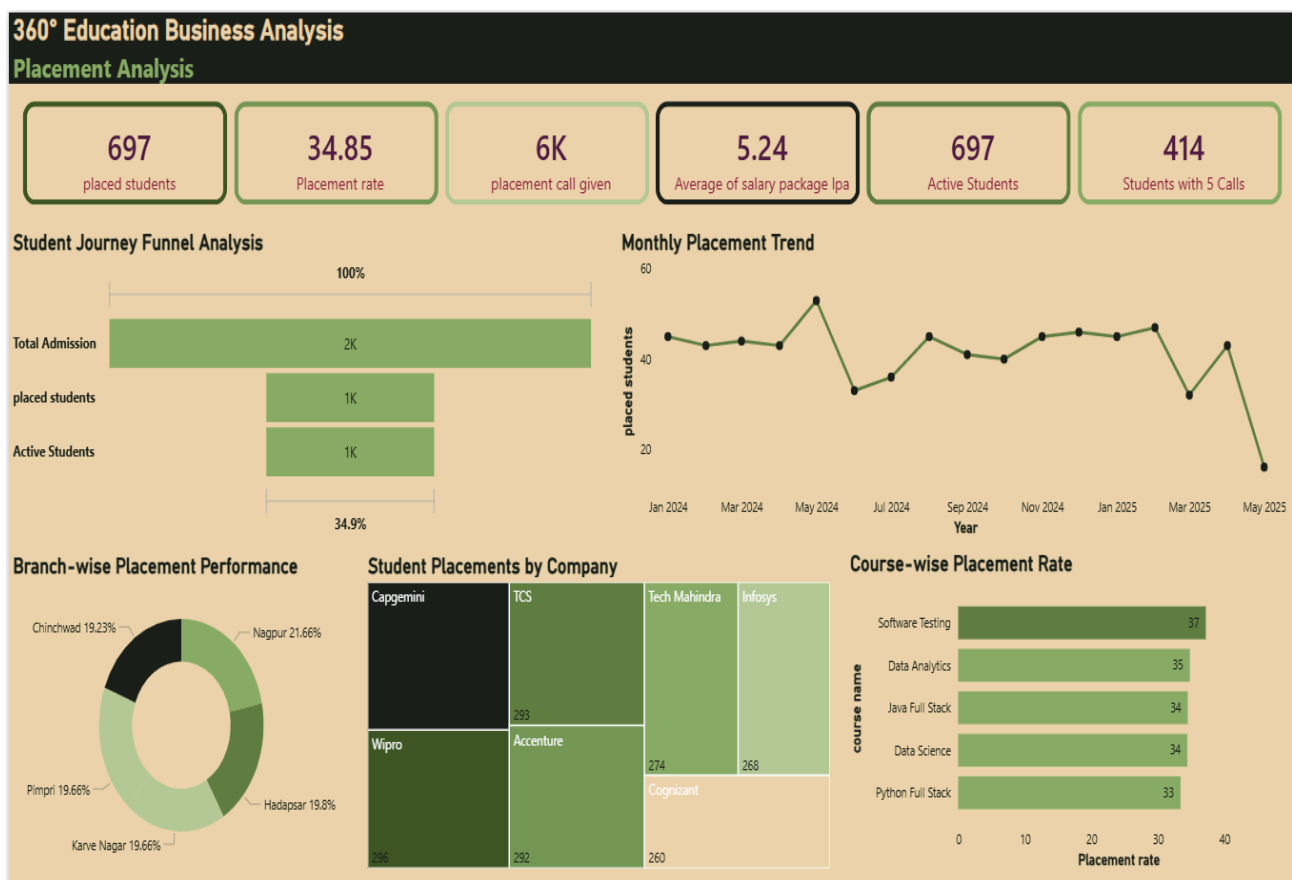
- **Key Metrics (KPIs):** Total Leads (2K), Leads Converted (638), Conversion % (31.90%), Effective Calls (11K), Total Calls (18K), and Average Revenue per Admission (33.03K).
- **Visualizations & Structure:**
 - **Sales Funnel:** Shows a standard drop-off, though the conversion metric notation seems structurally identical between total leads and total admissions (both listed as 2K, which indicates a data labeling overlap anomaly in the funnel visual).
 - **Lead Source Converter:** LinkedIn is the top-performing channel (123 converted), followed closely by Walk-Ins (105), YouTube (105), and Referrals (104). Website performance is trailing at 98.
 - **Branch Conversion & Comparison:** Displays total leads vs. converted leads per branch. Karve Nagar generated the highest total leads and conversions.



5.3 User & Usage Analytics Dashboard

This dashboard measures post-education success and student outcomes.

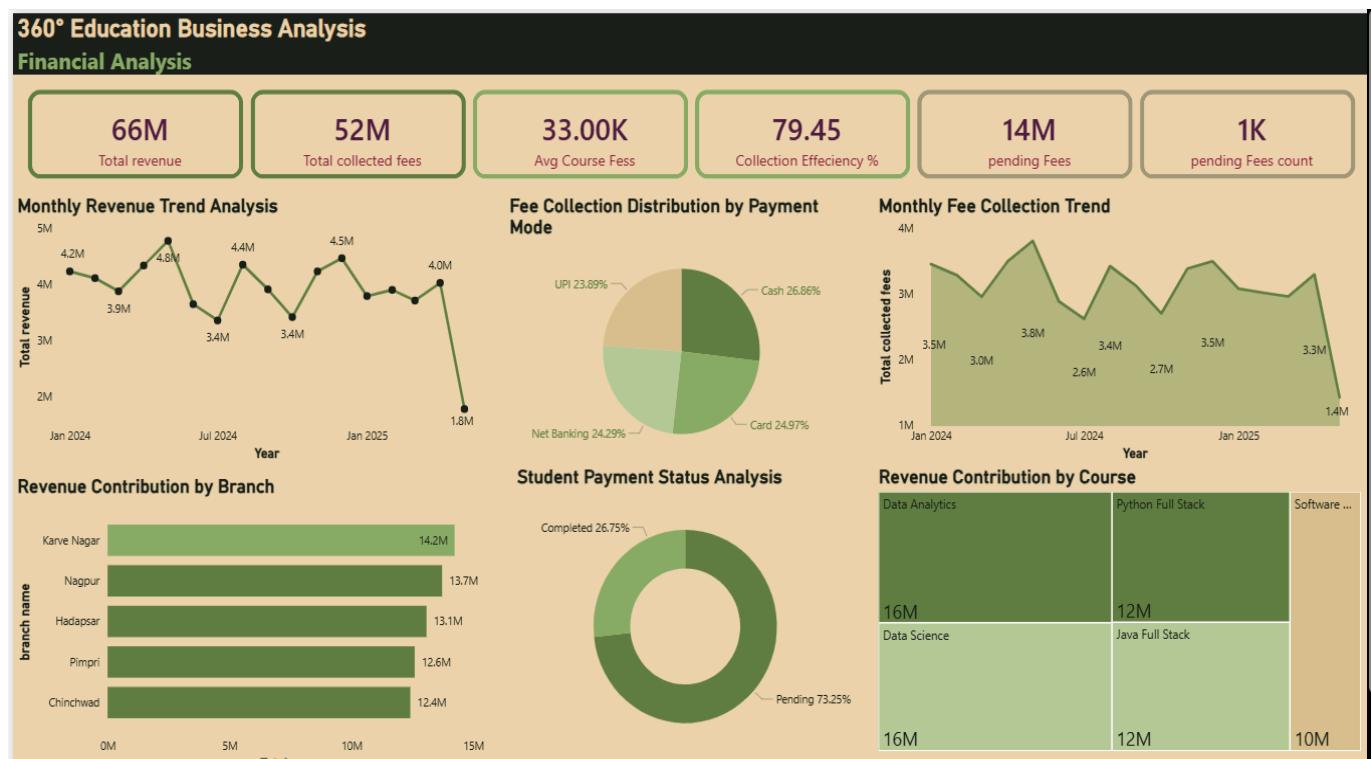
- **Key Metrics (KPIs):** Placed Students (697), Placement Rate (34.85%), Placement Calls Given (6K), Average Salary Package (5.24 LPA), Active Students (697), and Students with 5+ Calls (414).
- **Visualizations & Structure:**
 - **Monthly Placement Trend:** Solid, stable placement numbers hovering between 40-50 students monthly throughout 2024, followed by a sharp decline in mid-2025.
 - **Student Placements by Company:** Wipro (296), Capgemini (293), and TCS (293) are the primary corporate recruiters hiring from this institute.
 - **Course-wise Placement Rate:** Fairly uniform across courses, led by Software Testing (37%) and trailing slightly at Python Full Stack (33%).



5.4 Infrastructure & Server Monitoring Dashboard

This dashboard focuses on cash flow, collection efficiencies, and outstanding debts.

- **Key Metrics (KPIs):** Total Revenue (66M), Total Collected Fees (52M), Average Course Fees (33.00K), Collection Efficiency (79.45%), Pending Fees (14M), and Pending Fees Count (1K students).
- **Visualizations & Structure:**
 - **Student Payment Status Analysis:** A glaring 73.25% of students have pending fees, while only 26.75% have completed their payments fully.
 - **Fee Collection Distribution:** Payment modes are equally split among Cash (26.86%), Card (24.97%), Net Banking (24.29%), and UPI (23.89%).
 - **Revenue Contribution by Course:** Re-emphasizes that Data Analytics and Data Science bring in the lion's share of cash (16 M each).



6. Key Insights & Findings

- **The May 2025 Performance Collapse:** Across *all* timeline graphs (Revenue, Admissions, and Placements), there is a catastrophic drop-off in May 2025. This points to a macro operational block (e.g., incomplete data entry for the current month, a website breakdown, or a sudden halt in seasonal batch enrollments).
- **The Fee Collection Crisis:** While revenue looks strong at 66M, **14M is still outstanding**. Out of 2,000 students, 1,000 students (exactly half) owe pending fees. A collection efficiency of 79.45% is hazardous for an operational runway.
- **Low Placement Rate:** A placement rate of **34.85%** is low for a professional training institute. While 6,000 placement calls were given, only 697 students were successfully placed, showing a gap between corporate interview volume and student interview readiness.
- **Balanced Branch Performance:** There is no "failing" branch. All locations (Karve Nagar, Nagpur, Hadapsar, Pimpri, Chinchwad) hover remarkably close to one another in revenue (\$12.4\text{M}\$ to \$14.2\text{M}\$) and lead conversion.

7. Business Recommendations

- **Implement Aggressive Fee Tracking:** Since 1,000 students owe money, the institute should introduce milestone-locked learning. Access to advanced modules, certifications, or the placement cell should be revoked if pending fees are not cleared by mid-course intervals.
- **Revamp the Placement Bootcamps:** Since corporate tie-ups exist (TCS, Wipro, Capgemini are hiring in large batches) and 6,000 interview calls are being generated, the bottleneck is interview clearing. Introduce mandatory mock interviews, technical screening preparation, and soft skills workshops to boost the 34.85% placement rate.

- **Optimize Marketing Spend toward LinkedIn and YouTube:** LinkedIn, Walk-Ins, and YouTube produce the highest conversions. Marketing budgets should be diverted away from lower-performing channels and funneled directly into LinkedIn thought-leadership content and educational YouTube funnels.
- **Promote High-Value Courses:** Data Science and Data Analytics yield the highest financial returns. Double down on marketing these specific courses and consider bundling Software Testing (which has the highest placement rate) as an add-on or bridge program.

8. Conclusion

The business exhibits robust market demand, evidenced by a steady stream of 2,000 admissions, active corporate recruiting partnerships with Tier-1 companies, and a solid revenue foundation of 66M.

However, the institute faces two critical operational bottlenecks: a severe cash flow lag caused by 14M in uncollected fees from 50% of the student body, and an underperforming interview-to-placement pipeline. By formalizing strict fee collection milestones and enhancing student interview preparation, the company can quickly unlock trapped capital and boost its reputation through higher placement rates.

9. Future Improvements

- **Automated Fee Guardrails:** Link the learning platform to the payment backend to automatically pause course or placement access if fee milestones are missed.
- **AI-Powered Placement Prep:** Deploy automated resume-scanners and AI mock-interview tools to better prepare students *before* sending them to corporate interviews.

- **Predictive Lead Scoring:** Use CRM automation to score incoming leads, allowing sales teams to prioritize high-intent inquiries from LinkedIn and YouTube.
- **Real-Time Data Pipelines:** Upgrade the dashboards with automated, real-time data feeds to eliminate tracking lag and prevent artificial data drops (like the May 2025 anomaly).
- **Risk-Free Payment Options:** Partner with financial institutions to offer official EMI options or Income Share Agreements (ISAs), transferring fee-collection risks away from the business.